



## ANALYSTLAB AFRICA

### Data Analytics Internship Programme

Official Weekly Assignment

### Week 3: Advanced Data Analysis, KPI Development & Business Intelligence Dashboard

#### Document Control

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Programme           | Data Analytics Internship                                                 |
| Week                | Week 3                                                                    |
| Project             | Advanced Data Analysis, KPI Development & Business Intelligence Dashboard |
| Version             | 1.0                                                                       |
| Issue Date          | _____                                                                     |
| Submission Deadline | End of Week 3 (Sunday, 11:59 PM WAT)                                      |

#### 1. BUSINESS SCENARIO

You are continuing your role as a Data Analyst at AnalystLab Africa Consulting.

During the previous stages of your project, you worked with either the:

- Superstore Sales Dataset, or
- Global Superstore Dataset.

You explored the data, identified important patterns, analysed business performance, and developed insights relating to areas such as sales, profit, products, customers, regions, and business performance.

In Week 3, you will continue working with the same dataset you used for Week 2.

Your responsibility is to perform a deeper analysis of the data, develop meaningful Key Performance Indicators (KPIs), and create an interactive Business Intelligence Dashboard that communicates your findings clearly to decision-makers.

#### 2. IMPORTANT: DATASET CONTINUITY

You must continue working with the dataset you used in your Week 2 Data Analytics project.

If you used:

- **Superstore Sales Dataset** → Continue using the Superstore Sales Dataset.
- **Global Superstore Dataset** → Continue using the Global Superstore Dataset.

**Do not switch to the Telco Customer Churn Dataset.**

You are also not required to start a completely new project.

Your Week 3 analysis should build upon the work you have already completed.

### 3. OBJECTIVE

The objective of this assignment is to transform your previous analysis into a more advanced, business-focused project.

By the end of Week 3, you should be able to:

- Perform advanced data analysis.
- Identify important business trends and patterns.
- Develop meaningful KPIs.
- Perform time-based analysis.
- Analyse sales and profitability.
- Investigate product and customer performance.
- Identify underperforming areas of the business.
- Create calculated measures.
- Develop an interactive dashboard.
- Communicate business insights clearly.
- Provide data-driven recommendations.

### 4. TOOLS TO USE

#### Required Tools

You should use the following tools where applicable:

- Microsoft Power BI
- Microsoft Excel or Google Sheets
- GitHub
- Google Drive

#### Recommended Tools

You may also use:

- Power Query
- DAX
- Power BI Desktop

- Tableau Public
- SQL, where applicable

## Optional Tools

You may also use:

- Python
- Jupyter Notebook
- Google Colab

**Note:** You are not required to use every tool listed above. However, the final dashboard should be created using **Power BI or Tableau**.

## 5. ASSIGNMENT TASKS

### PART 1: REVIEW YOUR WEEK 2 ANALYSIS

Before beginning your Week 3 work, review the analysis you completed in Week 2.

Identify:

- The main business questions.
- Important findings.
- Key trends.
- High-performing products or categories.
- Low-performing products or categories.
- High-performing regions, markets, or locations.
- Low-performing regions, markets, or locations.
- Important customer segments.
- Profitability concerns.
- Questions that require deeper investigation.

You should use your Week 2 findings to guide your Week 3 analysis.

### Deliverable

Prepare a brief **Project Continuity Summary** explaining:

1. The dataset you are using.
2. The business problem.
3. The key findings from Week 2.
4. The areas you will investigate further in Week 3.

### PART 2: ADVANCED BUSINESS ANALYSIS

Conduct a deeper analysis of your dataset.

Your analysis should focus on meaningful business questions relevant to your selected dataset.

Possible areas of analysis include:

### **Sales Performance**

Investigate:

- Total Sales.
- Sales trends over time.
- Monthly sales performance.
- Quarterly performance, where applicable.
- Year-over-year growth, where applicable.
- High-performing and low-performing periods.

### **Profitability Analysis**

Investigate:

- Total Profit.
- Profit trends over time.
- Profit by category.
- Profit by sub-category.
- Profit by region or market.
- Products or categories generating losses.

### **Product Performance**

Analyse:

- Best-performing products.
- Worst-performing products.
- Best-performing categories.
- Low-performing categories.
- Products generating high sales but low profit.
- Products generating losses.

### **Regional or Geographic Performance**

Where applicable, investigate:

- Sales by region.
- Profit by region.
- High-performing locations.
- Low-performing locations.
- Regional growth opportunities.

For Global Superstore users, this may include:

- Market.

- Region.
- Country.
- City.

## Customer and Segment Analysis

Where applicable, investigate:

- Sales by customer segment.
- Profit by customer segment.
- High-value customers.
- Customer contribution to sales.
- Customer contribution to profit.

## Discount Analysis

Where discount information is available, investigate:

- The relationship between discounts and sales.
- The relationship between discounts and profit.
- Whether high discounts are associated with lower profitability.
- Categories or products that may be negatively affected by discounting.

## Deliverable

Document your **Advanced Business Analysis** in your Power BI report, notebook, or supporting analysis document.

## PART 3: KPI DEVELOPMENT

Develop meaningful **Key Performance Indicators (KPIs)** for your dashboard.

Your dashboard should include at least **six KPIs**.

Depending on your dataset, your KPIs may include:

- Total Sales.
- Total Profit.
- Total Orders.
- Total Customers.
- Average Sales per Order.
- Average Profit per Order.
- Profit Margin.
- Sales Growth.
- Profit Growth.
- Average Discount.
- Number of Loss-Making Products or Orders.

## Example Calculations

### Profit Margin

Profit Margin = Total Profit ÷ Total Sales

### Average Sales per Order

Average Sales per Order = Total Sales ÷ Number of Orders

You should ensure that every KPI included in your dashboard provides useful information for business decision-making.

## Deliverable

Create at least **six meaningful KPIs**.

## PART 4: DAX CALCULATIONS AND MEASURES

If you are using Power BI, create calculated measures using **DAX**.

You should create at least **five meaningful measures**.

Possible measures may include:

- Total Sales.
- Total Profit.
- Total Orders.
- Profit Margin.
- Average Sales per Order.
- Average Profit per Order.
- Total Discount.
- Sales Growth.
- Profit Growth.

## Example DAX Measure

Total Sales

Total Profit

Profit Margin

Note: You are not required to use these exact measures. Create measures that are appropriate for your dataset and analysis.

## Deliverable

Include at least **five DAX measures or equivalent calculated metrics**.

## PART 5: TIME-BASED ANALYSIS

Perform time-based analysis where your dataset contains date information.

Your analysis may include:

- Monthly sales trends.
- Monthly profit trends.
- Quarterly performance.
- Yearly performance.
- Sales growth.
- Profit growth.
- Seasonal patterns.

You should investigate questions such as:

- Which months perform best?
- Which months perform poorly?
- Are sales increasing or decreasing?
- Is profit following the same trend as sales?
- Are there seasonal patterns?

## Deliverable

Include meaningful **time-based visualisations and analysis**.

## PART 6: BUSINESS PROBLEM INVESTIGATION

Identify at least **three important business problems or opportunities** based on your analysis.

Examples may include:

### Problem 1: High Sales but Low Profit

A product or category may generate strong sales but produce limited profit.

Investigate possible reasons.

### Problem 2: Loss-Making Products

Identify products, sub-categories, or regions generating losses.

Investigate:

- Sales.
- Profit.
- Discounts.
- Order volume.

### Problem 3: Regional Underperformance

Identify regions, markets, countries, or locations that are underperforming.

Compare them with stronger-performing areas.

### Problem 4: Discount Impact

Investigate whether high discount levels are negatively affecting profitability.

### Problem 5: Customer or Segment Performance

Identify customer segments that contribute significantly to sales or profit.

### Deliverable

Identify and investigate at least **three meaningful business problems or opportunities**.

## PART 7: INTERACTIVE BUSINESS INTELLIGENCE DASHBOARD

Create an interactive dashboard that presents your analysis clearly.

Your dashboard should be designed for a business stakeholder or decision-maker.

### Dashboard Requirements

Your dashboard should include:

#### KPI Section

Include at least six key KPIs.

For example:

- Total Sales.
- Total Profit.
- Total Orders.
- Total Customers.
- Profit Margin.
- Average Sales per Order.



## Sales and Profit Analysis

Include visualisations showing:

- Sales trends.
- Profit trends.
- Sales by category.
- Profit by category.
- Regional performance.

## Product Performance

Include analysis of:

- Top-performing products.
- Low-performing products.
- Categories.
- Sub-categories.

## Customer or Segment Analysis

Where applicable, include:

- Customer segment performance.
- Top customers.
- Customer contribution.

## Geographic Analysis

Where applicable, include:

- Sales by region.
- Profit by region.
- Market or country performance.

## Filters and Interactivity

Include appropriate slicers or filters.

Examples include:

- Year.
- Region.
- Market.
- Category.
- Sub-category.
- Customer segment.

## Dashboard Design Requirements

Your dashboard should:

- Have a clear title.
- Use meaningful section headings.
- Be easy to understand.
- Avoid unnecessary clutter.
- Use appropriate visualisations.
- Include filters where relevant.
- Present information in a logical flow.

### Deliverable

Submit an **interactive Business Intelligence Dashboard**.

## PART 8: BUSINESS INSIGHTS AND RECOMMENDATIONS

Based on your analysis, identify at least **five important business insights**.

Your insights should answer meaningful questions such as:

- What is driving sales performance?
- What is affecting profitability?
- Which products or categories are underperforming?
- Which regions or markets provide growth opportunities?
- Are discounts affecting profitability?
- Which customer segments are most valuable?
- What actions should the organisation consider?

Your recommendations should be:

- Evidence-based.
- Practical.
- Clearly connected to your analysis.
- Relevant to the business problem.

### Deliverable

Prepare a **Business Insights and Recommendations Report**.

## 6. MINIMUM PROJECT REQUIREMENTS

Your Week 3 project must include:

- Advanced analysis building on Week 2.
- At least six meaningful KPIs.
- At least five calculated measures or DAX measures.

- Time-based analysis, where applicable.
- Product performance analysis.
- Sales and profitability analysis.
- Geographic or regional analysis, where applicable.
- Customer or segment analysis, where applicable.
- At least three investigated business problems or opportunities.
- At least five business insights.
- Evidence-based recommendations.
- An interactive Business Intelligence dashboard.

## 7. RECOMMENDED PROJECT STRUCTURE

You may organise your project as follows:

Continue using your existing project structure where possible.

## 8. DELIVERABLES

By the end of Week 3, submit:

1. Power BI Dashboard (.pbix) or approved BI tool project.
2. Dashboard PDF or screenshots.
3. Advanced Data Analysis document or notebook, where applicable.
4. Business Insights and Recommendations Report.
5. DAX Measures or Calculated Metrics documentation.
6. Updated README.md.
7. Updated GitHub Repository Link.
8. Google Drive Folder Link, where required.
9. Completed Week 3 Submission Form.

Ensure that all files and links are accessible before submission.

### Assessment Criteria

| Criteria                            | Weight |
|-------------------------------------|--------|
| Advanced Data Analysis              | 20%    |
| KPI Development                     | 15%    |
| DAX / Calculated Measures           | 15%    |
| Business Intelligence Dashboard     | 25%    |
| Business Insights & Recommendations | 15%    |
| Documentation & Professionalism     | 10%    |
| Total                               | 100%   |

### Professional Development Requirement

As part of your professional development, you should:

- Update your existing GitHub repository.
- Update your README.md.

- Document your analytical process.
- Present your dashboard professionally.
- Share your learning and project progress on LinkedIn and X where appropriate.
- Tag AnalystLab Africa where applicable.
- Use **#AnalystLabAfrica** where appropriate.

### Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.